Logic Circuits · AS · Practice

U03b · 24 written + 6 pen-first = 30 marks · Name: _____ Due: _____

Questions labelled Ours are authored. Preserve supplied expressions and circuit structure. NOT has one input; all other gates have two.

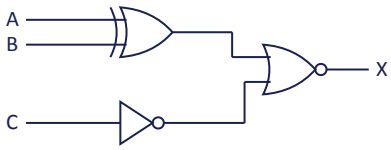
1. Ours: from the requirement

An aircraft warning horn H sounds if G=0 (landing gear not locked) while L=1 (low altitude), or if T=1 (test button pressed). Write the expression for H. **[2]**

2. Ours: expression into wires

Draw the circuit for **X = (A XOR B) NAND (B OR C)**. Label inputs and output; make connections clear. **[3]**

3. Ours: read the circuit



Write the logic expression for X, following the circuit shown. **[2]**

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4. Ours: follow every row

Complete the output column for **X = (A NAND B) AND (A OR C)**. Working columns are optional. **[4]**

A	B	C	A NAND B	A OR C	X
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

5. Ours: outputs back to an expression

For inputs AB in row order **00, 01, 10, 11**, the outputs are **1, 0, 0, 1**.

Write a correct expression using the six syllabus gates. No simplification is required. **[2]**

6. Ours: read the wiring

State what a bubble at a gate output means. State what a wire crossing with no junction dot means in these diagrams. **[2]**

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7. Ours: exam-shape rehearsal

New circuits, modelled on Cambridge drawing and table question shapes. These are authored questions, not past-paper extracts.

(a) Draw the circuit for **Z = (P OR Q) NOR (R NAND S)**. Preserve this structure; all gates have at most two inputs. **[2]**

(b) Complete the output column for **Z = (NOT P AND Q) OR (R XOR Q)**. **[2]**

P	Q	R	Working space	Z
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

8. Ours: test a proposed replacement

(a) Complete U = A XOR B and V = A OR B. **[4]**

A	B	U	V
0	0		
0	1		
1	0		
1	1		

(b) Give the input row that disproves U = V for all inputs. **[1]**

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9. Ours: match the fingerprint · pen first [6]

Paper first, then Circuit Forge. Two slots are wired as **X = (A ? B) ? C**. Choose AND, OR, XOR, NAND or NOR for each slot. The target outputs in row order **000 to 111** are **0, 0, 0, 1, 0, 1, 0, 1**. All six marks are human marked.

(a) Choose the two gates and sketch the circuit with its labels. **[2]**

(b) For your circuit, calculate X for inputs 001 and 110. Show the intermediate gate output in each case. **[2]**

(c) A classmate proposes (A AND B) OR C. Give one input combination where it differs from the target, and give both outputs. **[2]**

Keyboard companion: Karel 11 Spiral and 12 StoneMason remain the early-course programming tasks, outside these 30 marks. No Python file is needed for this gate investigation.

Above and beyond · unmarked, for both groups. A half-adder uses XOR for Sum and AND for Carry. Explain why two outputs are needed when adding 1+1. Why does a Boolean truth table alone not tell you how quickly a real adder responds?

Additional space - label the question number.